Precisa Surgical

Image-guided robotic drilling for minimally invasive ear and skull base surgery

Legal form: Srl (Milan, IT) **Founded:** 2021 **Employees:** 46

Stage: Series B (EUR 32 M, closed Q3 2026) **Contact:** Corporate Development, Precisa Surgical

1. Executive summary

Precisa Surgical builds **Precisa Drill**, a compact surgical robot that drills a tunnel of a few millimetres through the skull to reach the inner ear or the skull base, guided by the patient's CT scan. The first indication is cochlear implantation; later ones are skull base tumour biopsies and deep brain electrode placement.

We are looking for a **Swiss engineering and clinical research site** from 2027 and are evaluating Canton Bern for its surgical technology research and its university hospital.

What we are looking for in Bern

- Research groups in **image-guided surgery** and surgical navigation for ENT and neurosurgery.
- Expertise in **surgical robotics** and micromechatronic instruments.
- Partners in **hearing research** and otological surgery for cochlear implantation studies.
- **Medical 3D printing** of patient-specific anatomical models for surgical planning and training.
- **AI for medical image segmentation** of temporal bone CT scans.

2. Problem

- Cochlear implantation needs a large opening of the mastoid bone, takes 2–3 hours and needs highly trained surgeons.
- The facial nerve lies less than a millimetre from the drilling path; damage is rare but severe.
- Many hospitals cannot offer the procedure because of the learning curve.

3. Solution and technology

Precisa Drill

- Robot arm with sub-millimetre accuracy, fixed to the patient's head.
- Optical tracking and force sensing stop the drill near critical structures.
- Procedure time target: 45 minutes.

Precisa Plan

- Automatic segmentation of nerves and inner-ear structures from CT.
- Trajectory planning with safety margins.
- 3D-printed temporal bone models for surgeon training.

4. Market and business model

Revenue stream	Customer	Price
Precisa Drill system	University and regional hospitals	CHF 650,000 per system
Single-use drill kit	Hospitals	CHF 1,900 per procedure
Planning software and training models	Hospitals, ENT training centres	CHF 30,000 / year

About 80,000 cochlear implants are placed worldwide each year, growing about 8% per year.

5. Plans for Switzerland

- **Team:** 20 engineers and clinical specialists in Bern by 2029.
- **Space:** 500 m² of offices, an electronics lab and a mechanical workshop.
- **Regulatory path:** CE marking (MDR class IIb), first-in-human study planned for 2028.

6. Team

- **CEO:** co-founder, 15 years in surgical navigation companies.
- **CTO:** co-founder, PhD in medical robotics.
- **Medical director:** ENT surgeon specialised in cochlear implantation.

Fictional company and figures, created as a demo input for the HackApertus Bern Fit prototype. Any resemblance to real companies is coincidental.